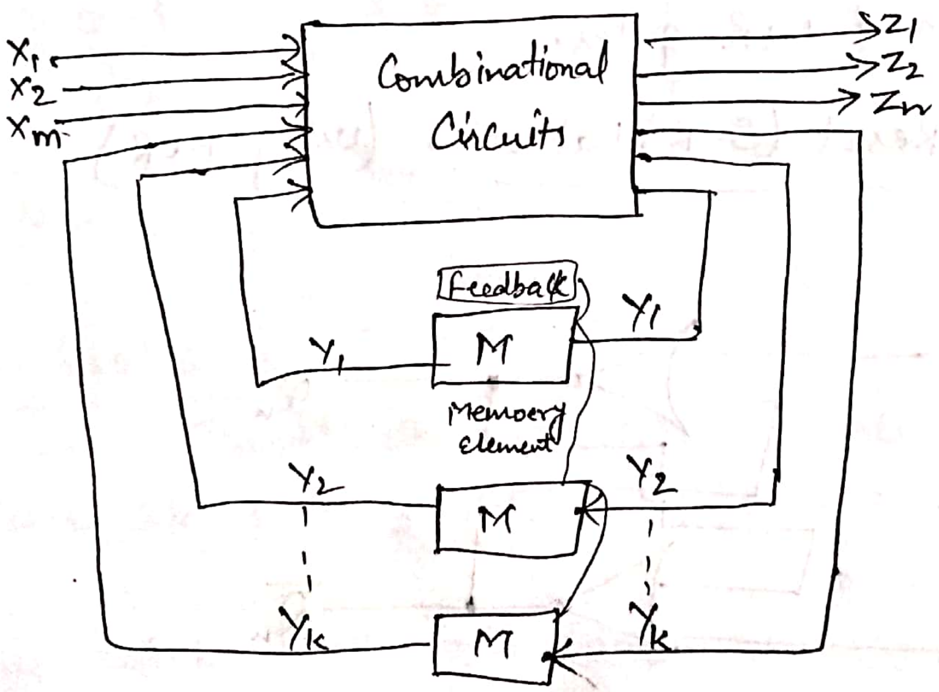
# Sequential Circuits :-



* The logic circuits whose o/p at any instant of time depend not only on the present i/p but also on the past o/p are called sequential circuits.

* In sequential circuits, the o/p signals are fedback to the i/p side.

* It consists of a combinational circuits to which memory elements are connected to ~~per~~ form a feedback path.



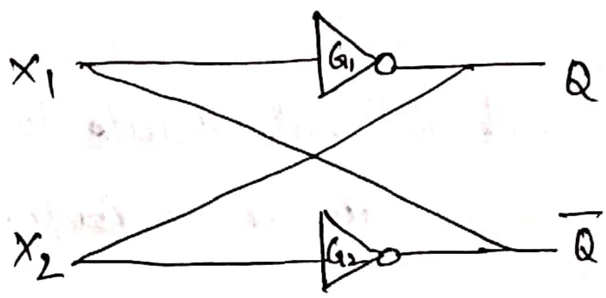

(Block diagram of sequential circuit)

* Sequential circuits are of two types-
  1. Synchronous or clocked.
  2. Asynchronous or unclocked.

① Latches :→

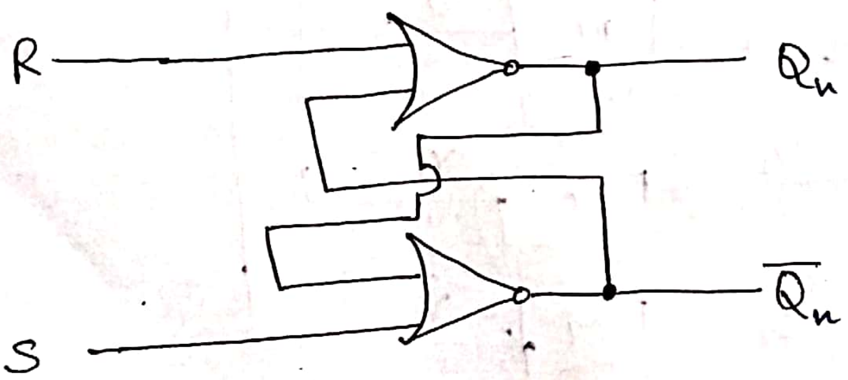
The simplest kind of sequential circuit has only two states. It is a memory cells which is capable of storing one bit information i.e. logic 0 or logic 1. This sequential circuit is called Latch.

*



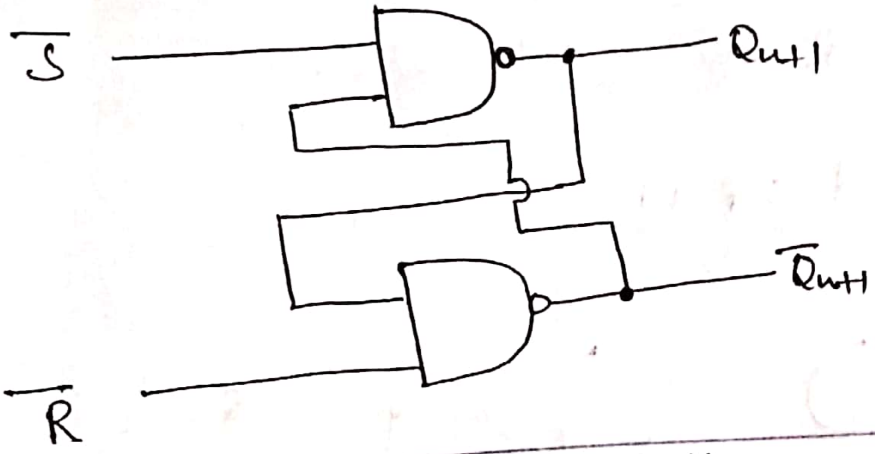

* $G_1$ and $G_2$ can be replaced with two i/p NAND and NOR gates.

(a) Set - Reset (S-R) Latch :→ (using NOR)




| I/P | | O/P | | Action |
|---|---|---|---|---|
| S | R | $Q_{n+1}$ | $\bar{Q}_{n+1}$ | |
| 0 | 0 | $Q_n$ | $\bar{Q}_n$ | No change |
| 0 | 1 | 0 | 1 | Reset |
| 1 | 0 | 1 | 0 | set |
| 1 | 1 | ? | ? | forbidden |

## NAND-based $\bar{S}$-$\bar{R}$ Latch :-




| I/P | | O/P | | Action |
|---|---|---|---|---|
| $\bar{S}$ | $\bar{R}$ | $Q_{n+1}$ | $\bar{Q}_{n+1}$ | |
| 0 | 0 | ? | ? | forbidden |
| 0 | 1 | 1 | 0 | Set |
| 1 | 0 | 0 | 1 | Reset |
| 1 | 1 | $Q_n$ | $\bar{Q}_n$ | No change. |

## Flip-flops :-

These are Synchronus Circuits, Change their states only when clock Pulses are Present.

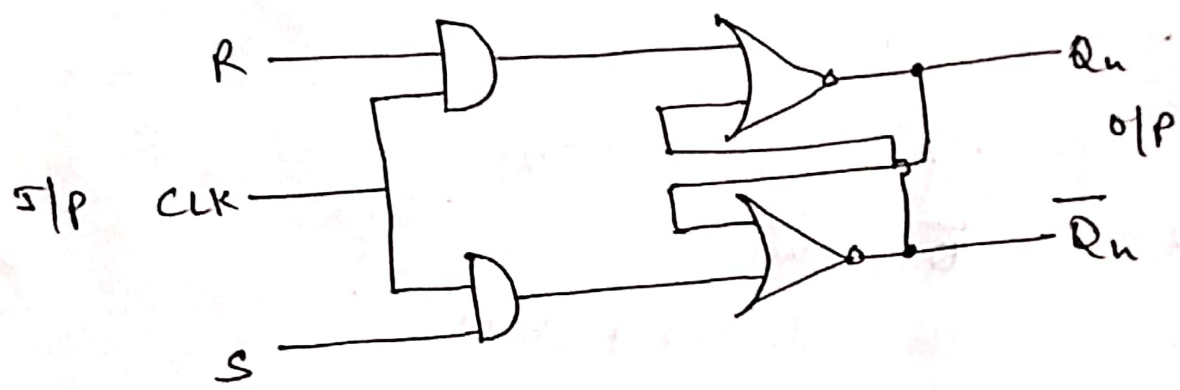
* The Latch with the additional Control I/P is called Flip-Flop.

* Flip-flops are of different types depending on how their i/ps and clock Pulses Cause transition b/w two states.

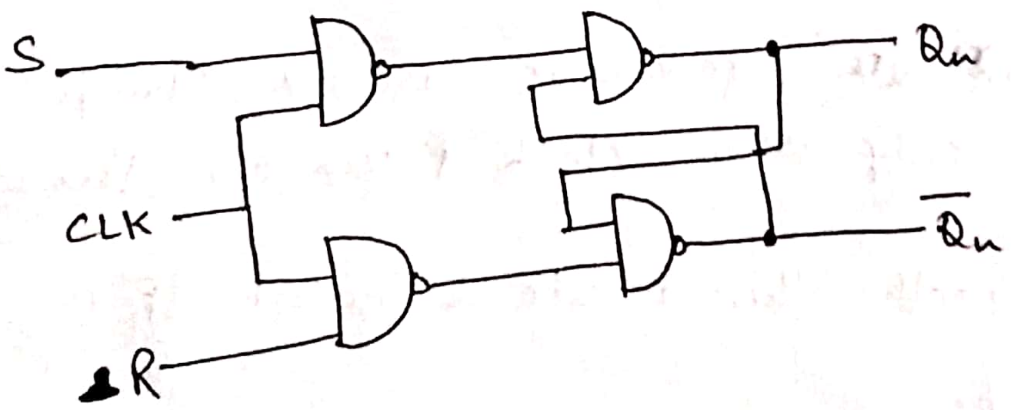
## ① S-R flip-flops :-

* The S-R ff Consists of two additional AND gates at the S-R I/Ps of S-R Latch.

* NOR Based S-R ff :-




* NAND Based S-R ff :-



## Excitation Table for SR flip flop :-

| PS $Q_n$ | NS $Q_{n+1}$ | Mendatory I/P S | R |
|---|---|---|---|
| 0 | 0 | 0 | X |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 1 | X | 0 |

X → 1/0
↓
dont care

## State eq^n. for S-R ff :-

### State table :-

| S | R | PS $Q_n$ | NS $Q_{n+1}$ |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 |
| 1 | 0 | 1 | 1 |
| 1 | 1 | 0 | X < 0, 1 |
| 1 | 1 | 1 | X < 0, 1 |

K-MAP for above table (To find $Q_{n+1}$)

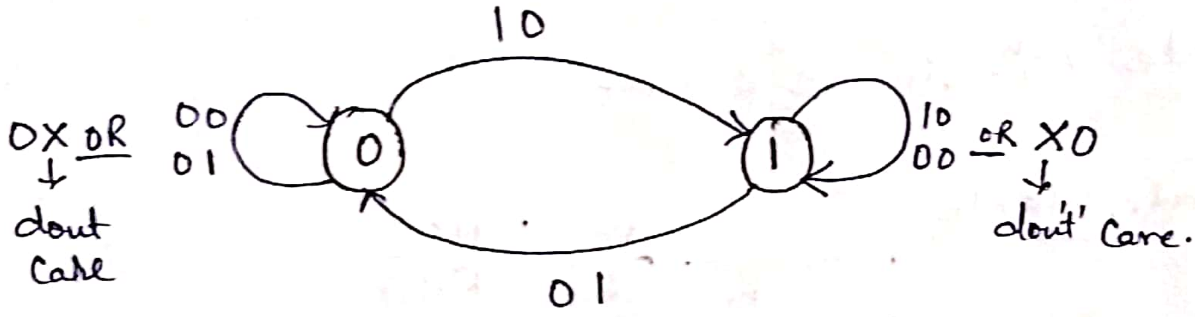
| Q \ SR | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | | | X | 1 |
| 1 | 1 | | X | 1 |

↓ $Q_n\bar{R}$ ↓ S

Now using K-map

$$\boxed{Q_{n+1} = S + Q_n\overline{R}}$$

## State diagram of S-R ff.



# Characteristics Table of S-R FF ⇒

OR State table ↴

| Present state $Q_n$ | CLK | Data I/Ps S | R | Next state $Q_{n+1}$ | Action |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | No change |
| 1 | 0 | 0 | 0 | 1 | No change |
| 0 | 1 | 0 | 0 | 0 | No change |
| 1 | 1 | 0 | 0 | 1 | No change |
| 0 | 0 | 0 | 1 | 0 | No change |
| 1 | 0 | 0 | 1 | 1 | No change |
| 0 | 1 | 0 | 1 | 0 | RESET |
| 1 | 1 | 0 | 1 | 0 | RESET |
| 0 | 0 | 1 | 0 | 0 | No Change |
| 1 | 0 | 1 | 0 | 1 | No Change |
| 0 | 1 | 1 | 0 | 1 | SET |
| 1 | 1 | 1 | 0 | 1 | SET |
| 0 | 0 | 1 | 1 | 0 | No change |
| 1 | 0 | 1 | 1 | 1 | No change |
| 0 | 1 | 1 | 1 | ? | forbidden |
| 1 | 1 | 1 | 1 | ? | forbidden |

i.e. Truth table

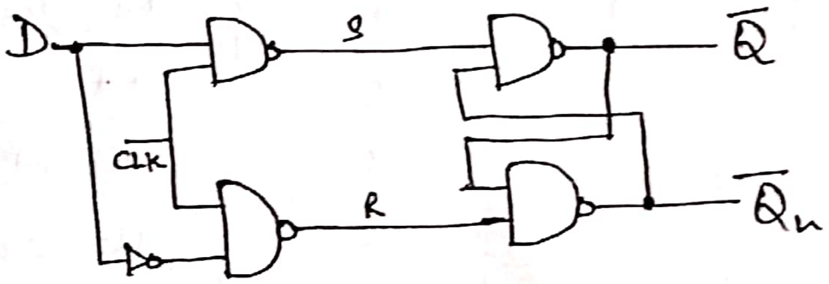
| S | R | Action |
|---|---|---|
| 0 | 0 | No Change |
| 0 | 1 | RESET |
| 1 | 0 | SET |
| 1 | 1 | forbidden |

← if CLK = 1

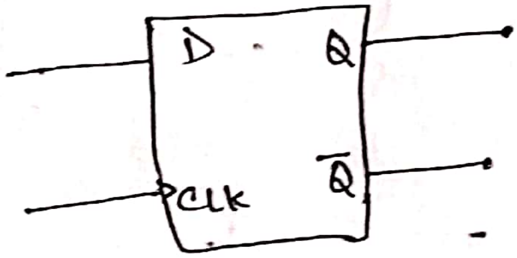
# D-flip-flop (Delay ff) :→

* The delay (D) flip-flop has only one input Called delay input and two outputs Q and $\overline{Q}$.

* It can be constructed from an S-R flip-flop by inserting and inverter b/w S and R and assigning the symbol D to the S-input.




(D ff using NAND gate)




(Logical symbol)

* When the CLK i/p is low (0) the D i/p has no effect.

* When CLK i/p is high (1), the Q o/p will taken on the value of D i/p.

* ~~When~~ if CLK = high (1) then if D = 0 then $Q_{n+1} = 0$
if D = 1 then $Q_{n+1} = 1$

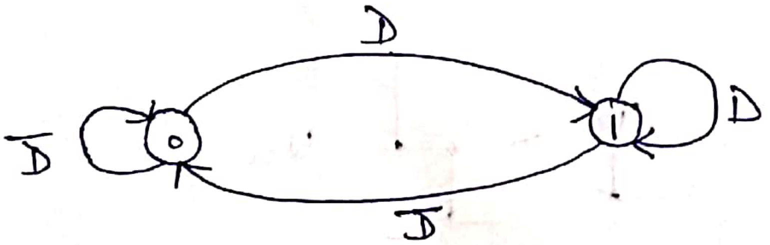
## Truth table of D ff :-

| CLK | I/P D | O/P $Q_{n+1}$ |
|---|---|---|
| 0 | X | No change |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

### Excitation Table

| $Q_n$ | $Q_{n+1}$ | D |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

## State diagram :-




## State table :-

| Present state $Q_n$ | Delay I/P D | Next state $Q_{n+1}$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

Using the state table The K-MAP for Next state $Q_{n+1}$ of D ff is - -

| $Q_n$ \ D | 0 | 1 |
|---|---|---|
| 0 | 0 | 1 |
| 1 | 0 | 1 |

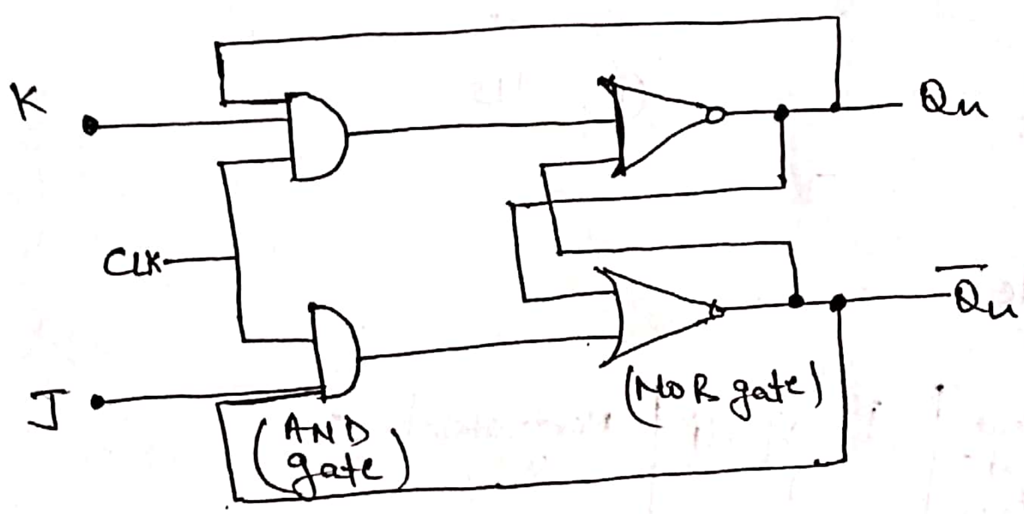
Characteristics eq^n.
OR
State equation.

i.e. $\boxed{Q_{n+1} = D}$

# J-K- flip-flop :-

* The Name JK flip-flop is termed the inventor Jack -kibly from texas instruments.
* The important feature of J-K ff as Compare to S-R flip-flop is toggling Nature.
* It is use in shift Register, PWM genreation etc.

## Circuit diagram :-




## Truth table :-

for CLK=1

| I/P | | | O/P |
|---|---|---|---|
| J | K | Qn | Qn+1 |
| 0 | 0 | X | No change (Qn) |
| 0 | 1 | X | 0 |
| 1 | 0 | X | 1 |
| 1 | 1 | X | Toggle |

for CLK = 0 in each Condition there is No change condition ~~of~~ obtain.

## ~~State table~~ of J-K ff.

Excitation Table.

| PS $Q_n$ | NS $Q_{n+1}$ | J | K |
|---|---|---|---|
| 0 | 0 | 0 | X |
| 0 | 1 | 1 | X |
| 1 | 0 | X | 1 |
| 1 | 1 | X | 0 |

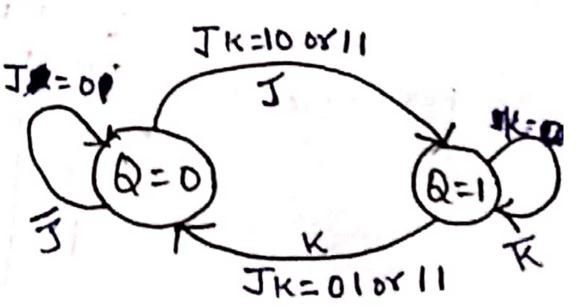
## Characteristic table

| J | K | $Q_n$ | $Q_{n+1}$ | Action |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | → No change |
| 0 | 0 | 1 | 1 | |
| 0 | 1 | 0 | 0 | → Reset |
| 0 | 1 | 1 | 0 | |
| 1 | 0 | 0 | 1 | → Set |
| 1 | 0 | 1 | 1 | |
| 1 | 1 | 0 | 1 | → Toggle. |
| 1 | 1 | 1 | 0 | |

K-MAP

| $Q_n$ \ JK | 00 | 10 | 11 | 10 |
|---|---|---|---|---|
| 0 | | | 1 | 1 |
| 1 | 1 | | | 1 |

state diagram :→

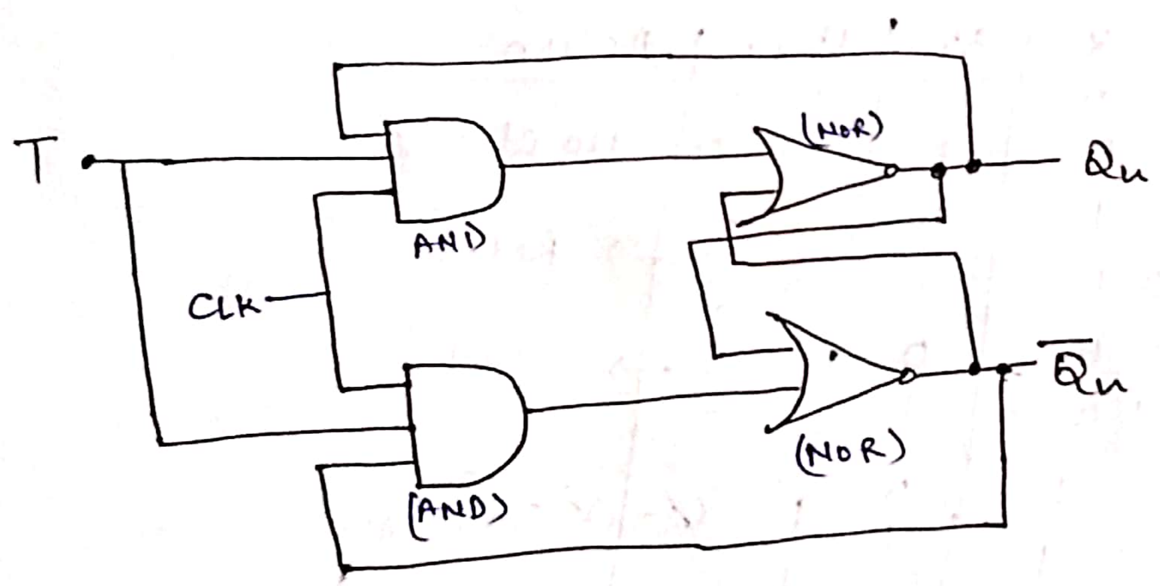



$$\boxed{Q_{n+1} = \overline{Q_n} J + Q_n \overline{K}}$$ ← State eqn.

## T- flip-flop :→ (Toggle flip-flop) :→

* The T or toggle flip-flop changes its output on each clock edge giving an output which is half the freq. of the signal to the T i/p.
* It is useful for constructing binary counters, freq. dividers and general binary addition devices.
* It can be made from a J-K flip-flop by tying both of its I/Ps high.

Circuit diagram :→




Truth table :→

For CLK=1

| I/P | | O/P | |
|---|---|---|---|
| T | $Q_n$ | $Q_{n+1}$ | |
| 0 | X | $Q_n$ | → No change. |
| 1 | X | $\overline{Q}_n$ | → Toggle. |

$$\boxed{Q_{n+1} = Q_n \oplus T}$$

For CLK=0, for each possible combination the o/p will be No change. i.e. $Q_{n+1} = Q_n$

## Characteristics table of T-flip-flop :-

| T | $Q_n$ | $Q_{n+1}$ | Action |
|---|---|---|---|
| 0 | 0 | 0 | No change |
| 0 | 1 | 1 | |
| 1 | 0 | 1 | Toggle |
| 1 | 1 | 0 | |

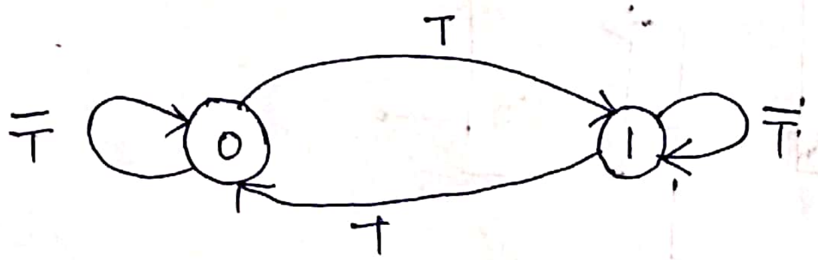
## K- MAP :-

| $Q_n$ \ T | 0 | 1 |
|---|---|---|
| 0 | | 1 |
| 1 | 1 | |

i.e $Q_{n+1} = Q_n \oplus T$

i.e $\boxed{Q_{n+1} = T\bar{Q}_n + \bar{T}Q_n}$

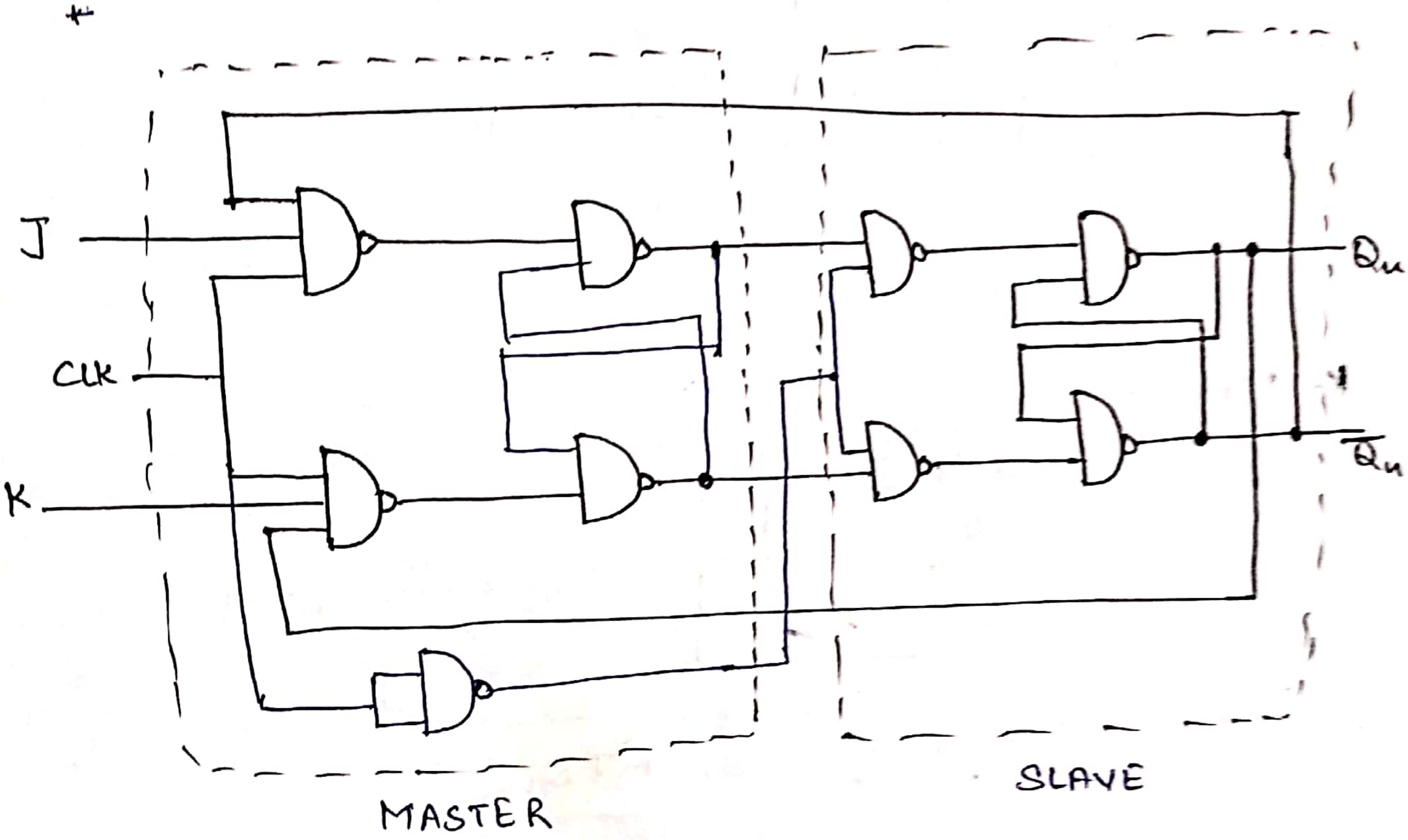
## State diagram :-



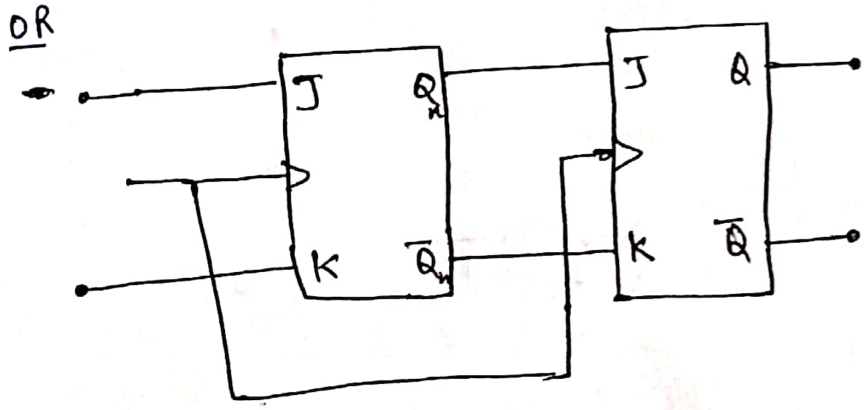
## J-K Master Slave flip-flop :→

A Master-slave flip-flop can be constructed using Two J-K flip-flops.

*




OR




* In the above diagram the first flip-flop called the MASTER is driven by the Positive edge of the clock (CLK) Pulse, the second flip-flop called the slave which is driven by Negative edge of the clock Pulse.

* When the Clock Pulse has +ve edge, the master acts according to its J-K inputs, but the slave does not respond since it requires a negative edge at the clock input.

* When the clock input has a negative edge, the Slave flip-flop copies the master outputs. But the master does not respond.

* Master-slave flip-flop does not have race-around problem.

* ① if J=1 and K=0, the master flip-flop sets on the +ve clock edge. The high $Q_n$ (1) o/p of the master drives the input (J) of the slave. So when the -ve clock edge hits, the slave also sets. Thus the slave FF copies the action of the master FF.

② If J=0 and K=1, the master resets on the +ve edge of the CLK. The high $\bar{Q}_n$ o/p of the master drives the i/p (K) of the slave flip-flop. Therefore when -ve clock edge hits the slave ff also RESETs.

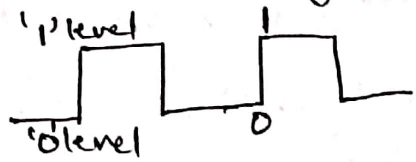
③ If J=K=1 then master ff toggles on the +ve CLK and the slave toggle on -ve clk.

④ If J=K=0 then No change Condition obtained.

# Triggering of ff :→

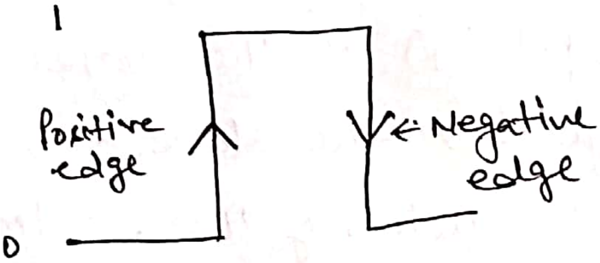
* Flip- Flops are Synchronous bistable devices.
* Term Synchronous means that the change in o/p occur at a specified Point on a triggering i/p called clock
* It can be classified into two different types

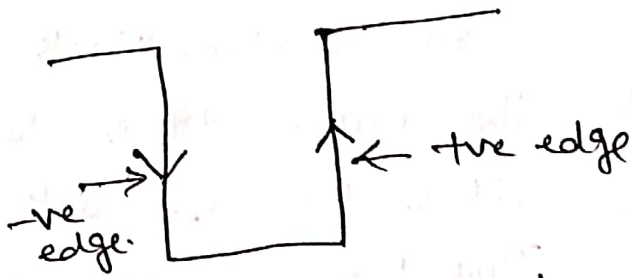
① Level triggering →




② Edge triggering

↓



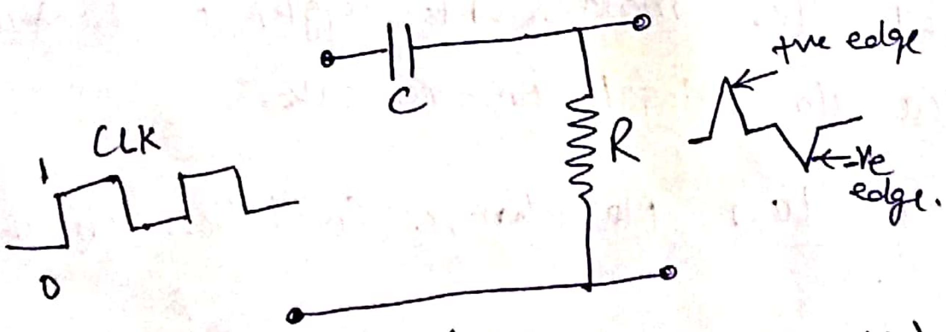




(Negative Pulse)

Positive Pulse

(~~+ve edge~~)

* +ve transition ($0 \xrightarrow{to} 1$) is +ve edge.
* -ve transition ($1 \xrightarrow{to} 0$) is -ve edge.



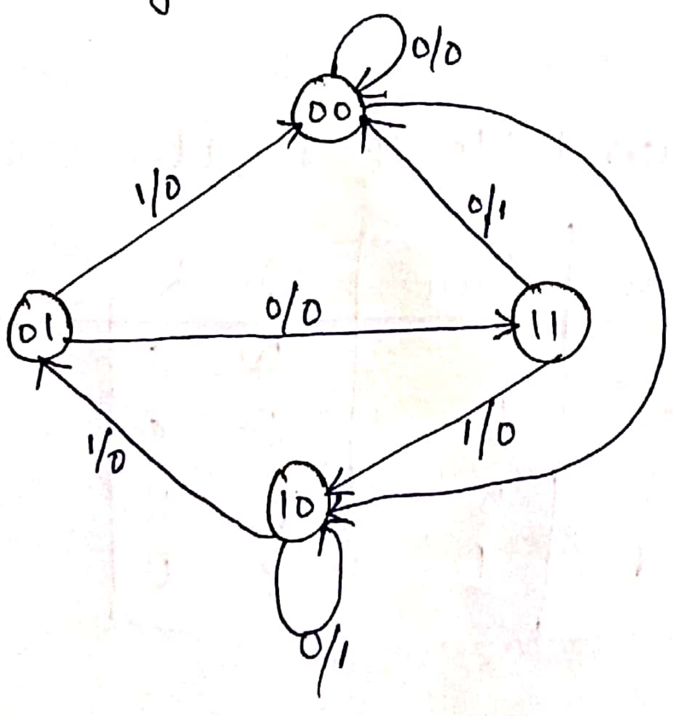

(differentiator Circuit)

# Synchronous Sequential Circuits :-

* The design Procedure of Synchronous sequential Circuits are

1. Determine State diagram.
2. from state diagram to State table.
3. State reduction (if applicable)
4. Binary assignment (if applicable) (if state diagram in alfabets)
5. By finding No. of flip-flop and assign letter to it.
6. Chose type of flip-flop (SR, JK, D, T)
7. Compare Excitation table and State table.
8. By K map find O/P of FF, I/P function
9. Draw the logic diagram.

Example :- A seq. circuit has one i/p and one o/p, state diagram shown below.
Design using JK ff.



② State table from State diagram

| PS | | Next State I/P | | | | O/P | |
|---|---|---|---|---|---|---|---|
| | | X = 0 | | X = 1 | | X = 0 | X = 1 |
| A | B | A | B | A | B | Y | Y |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 | 1 | 0 |

Step - 3 & 4 Not applicable ∵ we dont have any State reduction.

step ⑤ find No. of ff

$2^n \geq N$

$n \rightarrow$ No. of ff req.

$N \rightarrow$ No. of states

in this case

$2^n = 4$ $\boxed{\text{so} \Rightarrow n = 2}$ 2 ff req.

step-⑥ ff is JK

⑦ Excitation table of JK ff

| $Q_n$ | $Q_{n+1}$ | J | K |
|---|---|---|---|
| 0 | 0 | 0 | X |
| 0 | 1 | 1 | X |
| 1 | 0 | X | 1 |
| 1 | 1 | X | 0 |

**Step-7 →** Comparision of Excitation table and state table.

| PS | | I/P | NS | | FF I/P | | | | O/P |
|---|---|---|---|---|---|---|---|---|---|
| A | B | X | A | B | $J_A$ | $K_A$ | $J_B$ | $K_B$ | Y |
| 0 | 0 | 0 | 0 | 0 | 0 | X | 0 | X | 0 |
| 0 | 0 | 1 | 1 | 0 | 1 | X | 0 | X | 1 |
| 0 | 1 | 0 | 1 | 1 | 1 | X | X | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 0 | X | X | 1 | 0 |
| 1 | 0 | 0 | 1 | 0 | X | 0 | 0 | X | 1 |
| 1 | 0 | 1 | 0 | 1 | X | 1 | 1 | X | 0 |
| 1 | 1 | 0 | 0 | 0 | X | 1 | X | 1 | 1 |
| 1 | 1 | 1 | 1 | 0 | X | 0 | X | 1 | 0 |

**Step-8 →** K-MAP

for JA

| A \ BX | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | | 1 | | 1 |
| 1 | X | X | X | X |

Then $JA = B\overline{X} + \overline{B}X$

$= B \oplus X$

for KA

| A \ BX | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | X | X | X | X |
| 1 | | 1 | | 1 |

$KA = B\overline{X} + \overline{B}X$

$= B \oplus X$

for JB

| A \ BX | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | | | X | X |
| 1 | | 1 | X | X |

$J_B = AX$

for KB

| A \ BX | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | X | X | 1 | 0 |
| 1 | X | X | 1 | 1 |

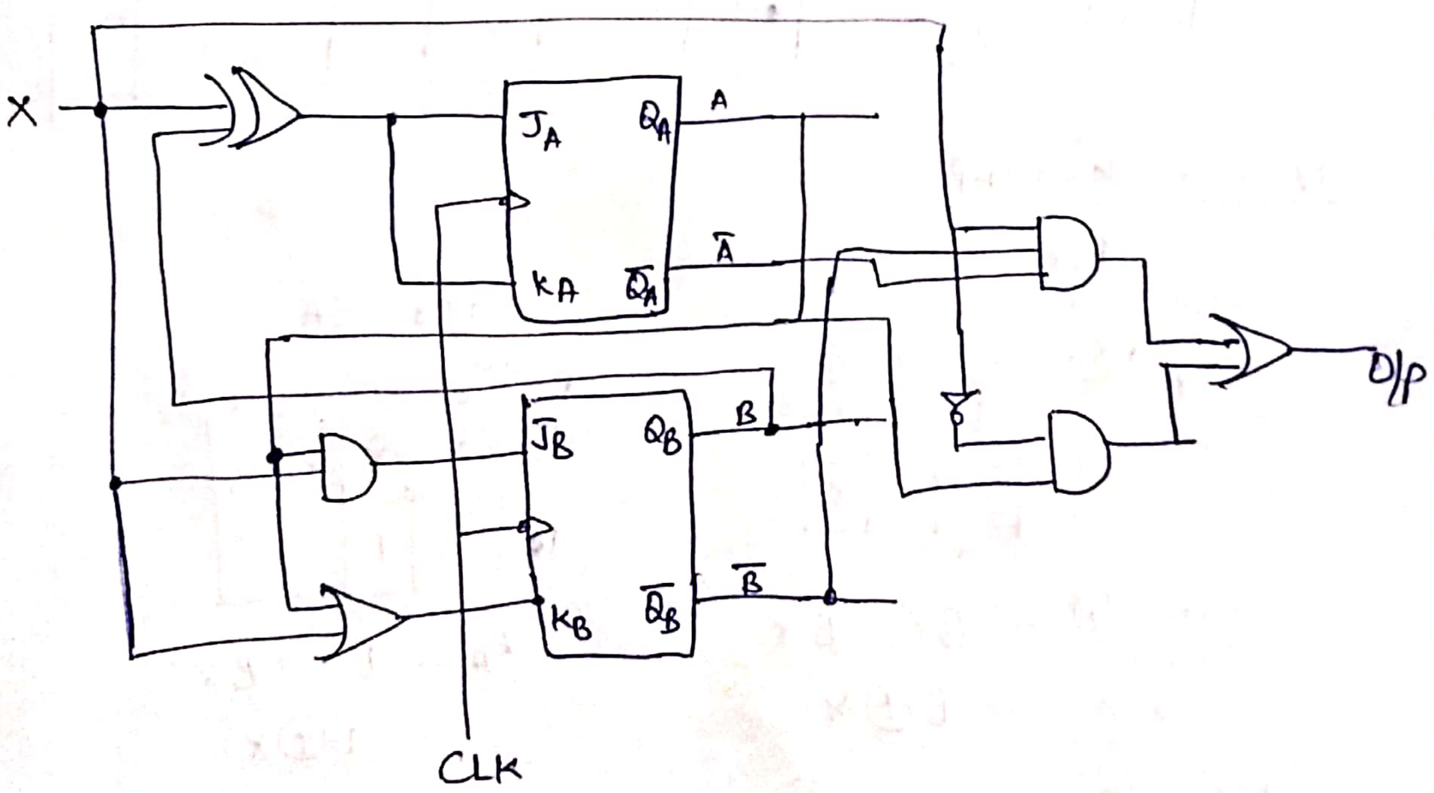
$K_B = A + X$

K-MAP for O/P is

| A \ BX | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | | 1 | | |
| 1 | 1 | | | 1 |

$$Y = \bar{A}\bar{B}X + A\bar{X}$$

Now logic diagram is



# Asynchronous Sequential Circuits :-

## Design Procedure :-

1. obtain a Primitive flow table
2. Reduce the flow table
3. Assign binary state Variable and obtain transition table.
4. Assign o/p values to dashes and obtain o/p maps.
5. Simplify the Excitation and o/p function.
6. Draw the logic diagram.

**Example:-** Design a gated latch Circuit with two inputs.

G → gate
D → Data

one o/p Q.

**sol^n :-** The gated latch is a memory element that accepts the value of D when G=1 and retains this value after G goes to 0.

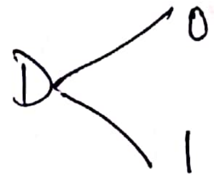
once G=0, a change in D does not change the value of o/p Q.

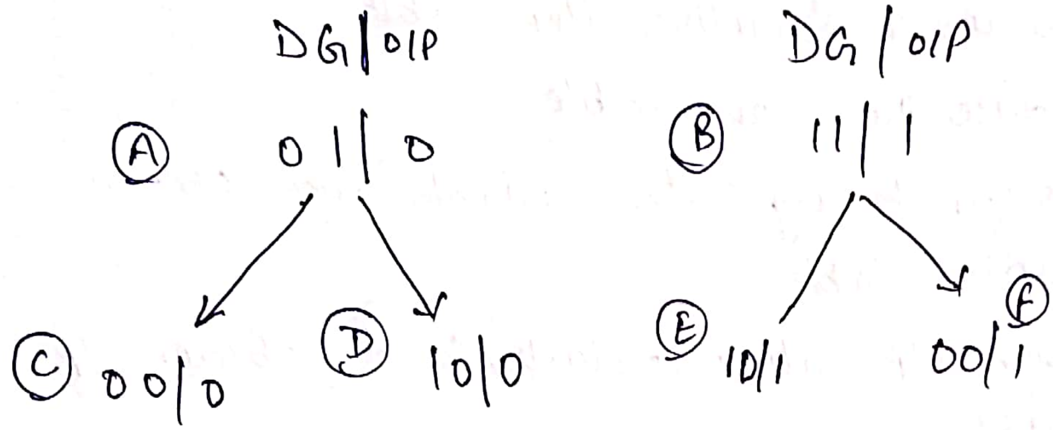
# if G value fixed to 1

D < 0, 1

| | DG | o/p |
|---|---|---|
| A | 01 | 0 |
| B | 11 | 1 |

if G value fixed to 0




i.e in table form

| State | I/P D | I/P G | O/P Q | Comment |
|---|---|---|---|---|
| A | 0 | 1 | 0 | → D = Q because G = 1 |
| B | 1 | 1 | 1 | → D = Q " " " |
| C | 0 | 0 | 0 | → After state A or D |
| D | 1 | 0 | 0 | → After state C |
| E | 1 | 0 | 1 | → " " D or F |
| F | 0 | 0 | 1 | → " " E |

## Flow table →

| | DG 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| A | C, – | A, 0 | B, – | –, – |
| B | –, – | A, – | B, – | E, – |
| C | C, 0 | A, – | –, – | D, – |
| D | C, – | [illegible], – | B, – | D, 0 |
| E | F, – | – – | B, – | E, 1 |
| F | F, 1 | A, – | –, – | E, – |

↳ dash
– → don't care condition

④ ~~Reduction~~ Implication table is

using flow-table we will design it.

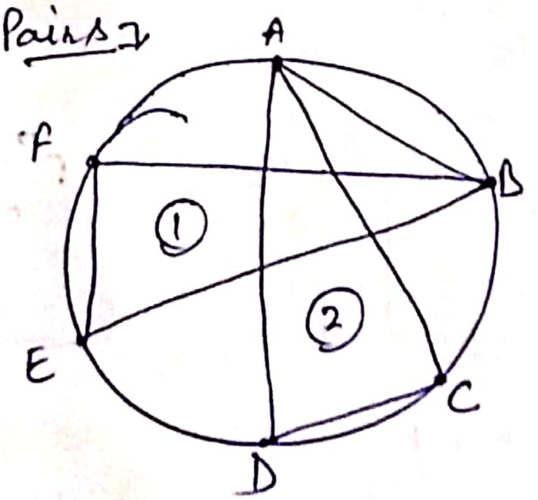
leave A in vertical and F in Horizontal case.

| | A | B | C | D | E |
|---|---|---|---|---|---|
| B | ✓ | | | | |
| C | ✓ | D,E X | | | |
| D | ✓ | D,E X | ✓ | | |
| E | C,F X | ✓ | DE C,F X | X | |
| F | C,F X | ✓ | X | D,E,X CF | ✓ |

Compatible Pairs

(A,B) . (A,C). (A,D) (B,E) , (B,F)
(C,D) , (E,F)

using above Pairs I




Now closed Paths are
① A C D
② B E F

Now using Closed Paths

| | DG 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| A | C,- | A,0 | B,- | -,- |
| C | C,0 | A,- | -,- | D,- |
| D | C,- | -,- | B,- | D,0 |

| | DG 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| B | -,- | A,- | B,1 | E,- |
| E | F,- | -,- | B,- | E,1 |
| F | F,1 | A,- | ~~-,-~~ | E,- |

Same as previous table.

State that are merging

| | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| ACD | C,0 | A,0 | B,- | D,0 |
| BEF | F,1 | A,- | B,1 | E,1 |

| | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| A | A,0 | A,0 | B,- | A,0 |
| B | B,1 | A,- | B,1 | B,1 |

A,C,D = A
B,E,F = B

this is reduced flow table.

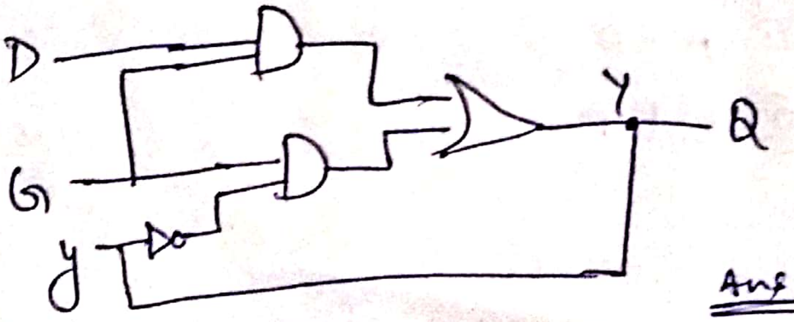
Transition table :-

| y | DG 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 |
| 1 | 1 | 0 | 1 | 1 |

| y | 00 | 01 | 11 | 10 |
|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 |

$$y = DG + G\bar{y}$$

$$Q = y$$




Ans